

Linear Inequalities — 30 MCQs

Digital SAT Style | Easy / Medium / Hard

Q1. DIFFICULTY: EASY

Which of the following is a solution to the inequality $4x - 3 > 9$?

- A) 1
 - B) 2
 - C) 3
 - D) 4
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Q2. DIFFICULTY: EASY

A student must score at least 70 points to pass a test. The student has already earned 48 points. If each remaining question is worth 4 points, what is the minimum number of additional questions the student must answer correctly to pass?

- A) 4
 - B) 5
 - C) 6
 - D) 7
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Q3. DIFFICULTY: EASY

Which value of x satisfies $2x + 5 \leq 13$?

- A) 3
 - B) 5
 - C) 6
 - D) 7
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Q4. DIFFICULTY: EASY

On a number line, the solution to $-3x < 12$ is best represented by which of the following?

- A) $x < -4$
 - B) $x > -4$
 - C) $x < 4$
 - D) $x > 4$
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Q5. DIFFICULTY: EASY

A company rents vans for a flat fee of \$80 plus \$0.25 per mile driven. Which inequality represents the situation where the total rental cost is no more than \$200?

- A) $80 + 0.25m < 200$
- B) $80 + 0.25m \leq 200$

C) $80 + 0.25m > 200$

D) $80 + 0.25m \geq 200$

Q6. DIFFICULTY: EASY

If $-4 \leq 2x \leq 10$, what is one possible integer value of x ?

GRID-IN

[GRID-IN — enter your answer in the box]

Q7. DIFFICULTY: EASY

A store sells bottles of water for \$1.50 each. Nadia wants to spend less than \$12. Which inequality represents the number of bottles b she can buy?

A) $1.5b < 12$

B) $1.5b \leq 12$

C) $1.5b > 12$

D) $b/1.5 < 12$

Q8. DIFFICULTY: EASY

What is the largest integer value of x that satisfies $5x - 1 < 19$?

A) 3

B) 4

C) 5

D) 6

Q9. DIFFICULTY: EASY

What is the smallest integer value of x that satisfies $3(x + 4) \geq 21$?

A) 1

B) 2

C) 3

D) 4

Q10. DIFFICULTY: EASY

A number line shows the solution set with an open dot at 5 and shading extending to the left. Which inequality describes this solution set?

A) $x \geq 5$

B) $x > 5$

C) $x \leq 5$

D) $x < 5$

Q11. DIFFICULTY: MEDIUM

For what values of x is $2x - 7 < 3x + 1$ AND $x - 4 \leq 8$ both satisfied simultaneously?

- A) $-8 < x \leq 12$
 - B) $x > -8$
 - C) $x \leq 12$
 - D) $-8 \leq x \leq 12$
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Q12. DIFFICULTY: MEDIUM

In the xy -plane, the solution set is the shaded region above a dashed line with equation $y = (2/3)x - 4$. Which inequality is represented?

- A) $y \leq (2/3)x - 4$
 - B) $y < (2/3)x - 4$
 - C) $y \geq (2/3)x - 4$
 - D) $y > (2/3)x - 4$
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Q13. DIFFICULTY: MEDIUM

If $2 \leq p \leq 8$ and $-3 \leq q \leq 2$, what is the maximum possible value of $p - q$?

- A) 5
 - B) 8
 - C) 10
 - D) 11
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Q14. DIFFICULTY: MEDIUM

A nutritionist recommends no more than 2,400 calories per day. A breakfast contains 520 calories and lunch contains 680 calories. If each serving of the dinner entrée contains 320 calories, what is the maximum number of dinner servings the person can eat?

- A) 2
 - B) 3
 - C) 4
 - D) 5
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Q15. DIFFICULTY: MEDIUM

Which of the following is the solution set to $|2x - 6| \leq 8$?

- A) $-1 \leq x \leq 7$
 - B) $x \leq -1$ or $x \geq 7$
 - C) $-7 \leq x \leq 1$
 - D) $x \leq 7$
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Q16. DIFFICULTY: MEDIUM

In the xy -plane, the boundary line of an inequality $rx + ty \geq -6$ passes through $(0, -2)$ and $(3, 0)$. What is the value of $r + t$?

GRID-IN

[GRID-IN — enter your answer in the box]

Q17. DIFFICULTY: MEDIUM

A contractor charges \$250 for materials and \$65 per hour of labor. A client's total budget is at most \$900. What is the maximum number of full hours of labor the contractor can work?

- A) 8
 - B) 9
 - C) 10
 - D) 11
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Q18. DIFFICULTY: MEDIUM

Which of the following values of y is NOT a solution to the inequality $3y - 2 > 4y + 1$?

- A) -4
 - B) -5
 - C) -6
 - D) -3
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Q19. DIFFICULTY: MEDIUM

The graph of $y > -(3/4)x + 5$ is a shaded half-plane. Which ordered pair lies in the solution region?

- A) $(4, 2)$
 - B) $(0, 3)$
 - C) $(0, 4)$
 - D) $(0, 6)$
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Q20. DIFFICULTY: MEDIUM

For what positive integer values of n does $2n - 1 < 15$ AND $n^2 > 9$?

GRID-IN (enter the number of such values)

[GRID-IN — enter your answer in the box]

Q21. DIFFICULTY: MEDIUM

A gym requires that members be at least 16 years old to enter without a signed parental consent form. Which inequality describes all ages a (in years) that allow entry without a consent form?

- A) $a < 16$

- B) $a \leq 16$
 - C) $a \geq 16$
 - D) $a > 16$
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Q22. DIFFICULTY: MEDIUM

If $-5 \leq m \leq 4$ and $-2 \leq n \leq 6$, what is the minimum possible value of mn ?

- A) -30
 - B) -20
 - C) -10
 - D) -8
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Q23. DIFFICULTY: HARD

The system of inequalities is graphed in the xy -plane.

$$y \geq x + 1$$

$$y \leq -2x + 4$$

Which quadrant(s) contain NO part of the solution region?

- A) Quadrant II only
 - B) Quadrant IV only
 - C) Quadrants III and IV
 - D) Quadrant III only
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Q24. DIFFICULTY: HARD

For what value of the constant k does the inequality $kx + 4 > 2x + k$ have solutions for ALL real numbers x ?

- A) 0
 - B) 2
 - C) 4
 - D) 6
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Q25. DIFFICULTY: HARD

A factory produces two products, X and Y. Each unit of X requires 3 hours of machine time and each unit of Y requires 5 hours. The machine can run at most 60 hours per week. The factory must produce at least 4 units of Y per week. Which system of inequalities represents these constraints?

- A) $3x + 5y \leq 60, y \geq 4, x \geq 0$
 - B) $3x + 5y \geq 60, y \leq 4, x \geq 0$
 - C) $5x + 3y \leq 60, y \geq 4, x \geq 0$
 - D) $3x + 5y \leq 60, y \leq 4, x \geq 0$
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Q26. DIFFICULTY: HARD

In the xy -plane, the boundary line passes through $(-2, 7)$ and $(4, -5)$. The solution region is below the line with a solid boundary. Which inequality represents this?

- A) $y \leq -2x + 3$
 - B) $y \leq 2x + 3$
 - C) $y \leq -2x - 3$
 - D) $y < -2x + 3$
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Q27. DIFFICULTY: HARD

If $k > 0$ and $4 - kx \geq 2 - 3x$, which gives all values of x that satisfy the inequality?

- A) $x \leq 2/(k-3)$ when $k > 3$; $x \geq 2/(k-3)$ when $0 < k < 3$
 - B) $x \leq 2/(k-3)$ for all $k > 0$
 - C) $x \geq 2/(k-3)$ for all $k > 0$
 - D) $x \leq 2$ for all $k > 0$
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Q28. DIFFICULTY: HARD

A candidate needs more than 60% of 10,000 votes to win outright. After 4,000 votes are counted, the candidate has 2,600. At minimum, how many of the remaining 6,000 votes must the candidate receive?

GRID-IN

[GRID-IN — enter your answer in the box]

Q29. DIFFICULTY: HARD

Which of the following inequalities has the same solution set as $-3(2x - 1) \leq 6x - 9$?

- A) $x \geq 1$
 - B) $x \leq 1$
 - C) $x \geq 2$
 - D) $x \leq -1$
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Q30. DIFFICULTY: HARD

In the xy -plane, point (a, b) is a solution to both $y \geq 3x - 1$ and $y < -x + 7$. Which of the following must be true?

- A) $a < 2$
 - B) $b > 5$
 - C) $a > 2$
 - D) $b < 5$
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